

Carbon → Google

PYTHON

Fundamentals, Data Manipulation

Dart

Flutter

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UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Outline

- 1 Foundations of Python
- 2 Numerical Analysis with Numpy
- 3 Data Manipulation with Pandas



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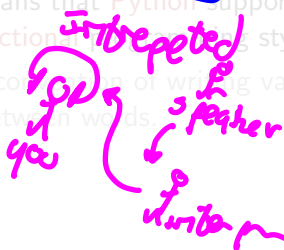
Basic Definitions in Python

→ 1993 (4) → Guido Van Rossum

- **Python** is a **high-level**, **interpreted**, and **general-purpose** programming language.
- **High-level** means that **Python** is designed to be easy to read and write.

Mathematical Notation
Natural Language

- **Interpreted** means that Python code is executed line by line, rather than being compiled into machine code.
- **Weakly typed** means that Python does not require you to declare the type of a variable.
- **Multiparadigm** means that Python supports object-oriented, imperative, and functional programming styles.



Basic Definitions in Python

1st x = 5 / Java
C

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- **Snake-case** is the convention of writing variable names in **lowercase**, with **underscores** between words.

Python → C
x = 9
x = "hello world"
x = False
cpython

~ Py code
↳ x.pyc



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Basic Definitions in Python

Camel Case } *my Variable Name*

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✓ my_name_variable



Popular Developer Tools

Life is Short, Use Dev Tools ByteByteGo

Development Env

- VS Code
- Visual Studio
- IntelliJ IDEA
- Notepad++
- Vim
- PyCharm
- Jupyter Notebook

Diagramming

- draw.io
- excalidraw
- mindmap
- Mermaid
- PlantUML
- Microsoft Visio
- Miro

AI Tools

- ChatGPT
- GitHub Copilot
- Tabnine
- Claude
- ollama
- Midjourney
- Stable Diffusion

Hosting & Deployment

- AWS
- Cloudflare
- Git Hub
- Fly.io
- Heroku
- Digital Ocean

Code Quality

- Jest
- ESLint
- Selenium
- sonarQube
- FindBugs
- Checkstyle

Security

- 1Password
- LastPass
- OWASP
- snyk
- Nmap

Note-taking

- Notion
- Markdown
- Obsidian
- Roam
- Logseq
- Tiddly Wiki

Design

- Figma
- Sketch
- Adobe Illustrator
- Canva
- Adobe Photoshop

Handwritten Annotations:

- Jet Brains** (circled around VS Code)
- PyTest** (circled around Jest)
- PyLint** (circled around ESLint)
- GCP** (circled around AWS)



Virtual Environments

- **Virtual environments** are a way to create isolated spaces on your computer for **Python projects**.
- **Virtual environments** allow you to install packages and dependencies for a specific project without affecting other projects.

Python 2

Python 3

Python 2
Numpy 1.4
Keras 2.5

Python 3
Numpy 2.4
Keras 1.7



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- **Virtual environments** are created using the **venv** module, which is included in the Python standard library.
- **Virtual environments** are activated using the **source** command in the terminal.
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Modules and Packages

- **Modules** are Python files that contain functions, classes, and variables.
- **Modules** are used to organize code and make it reusable.

• Packages are directories that contain Python files (modules).

• Packages are used to organize modules and make them reusable.

test.py

main.py



Modules and Packages

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- **Modules** are used to organize code and make it reusable.
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Handwritten annotations:

- A blue arrow points from the word "Packages" in the third bullet point to the text "Java".
- A blue arrow points from the word "modules" in the fourth bullet point to the text "--init--py".
- Blue circles highlight the words "Packages" and "directories" in the third bullet point.
- A blue bracket underlines the phrase "organize modules and make them reusable" in the fourth bullet point.



Dependencies Management

- **Dependencies** are external libraries or packages that your Python project relies on.
 - **Dependencies** are traditionally managed using a pyproject.toml file, which lists the names and versions of the packages required by your project.
 - **Dependencies** can be installed using the pip package manager, which is included with Python.
 - Currently, the poetry package manager is recommended for managing dependencies in Python projects. It follows the PEP 508 standard for specifying dependencies.
- avoid conflicts



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Jupyter Notebooks

Data Analysis

- **Jupyter Notebooks** are a web-based interactive computing environment that allows you to create and share documents that contain live code, equations, visualizations, and narrative text.

Jupyter Notebooks are used for data cleaning, data transformation, statistical modeling, data visualization, machine learning, and more.

- Jupyter Notebooks support multiple programming languages, including Python, R, and Julia.
- Jupyter Notebooks are used by data scientists, researchers, students, and educators to explore data, create reports, and teach programming.

education { Carol Willing
Fernando Pérez



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basic

classic



Jupyter Notebooks

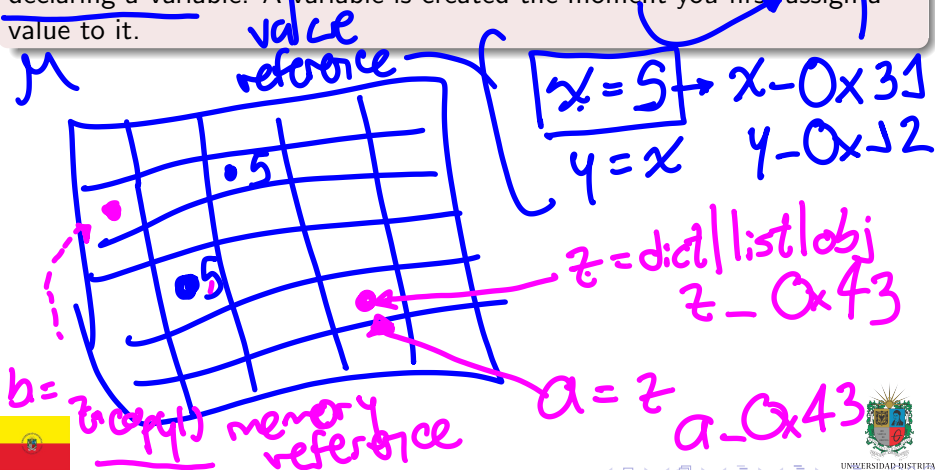
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Variables and Memory Management

Variables Definition

Variables are used to store data values. Python has no command for declaring a variable. A variable is created the moment you first assign a value to it.



Conditionals

Definition

Conditionals are used to execute different code blocks based on *different conditions*.

Nested Conditionals

Nested conditionals are conditionals that are used with other conditionals.

Elif Conditionals

Elif conditionals are used to execute multiple conditions.

if condition:

tab

else:

tab



Conditionals

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Conditionals are used to execute different code blocks based on *different conditions*.

Nested Conditionals

Nested conditionals are conditionals that are *inside* other conditionals.

Elif Conditionals

Elif conditionals are used to check *multiple conditions*.

if cond_1:

.... if cond_2:

.... if cond_3:



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(switch)

else if
elif

if cond-1:
elif cond-2:
elif cond-3:
else:



Loops and Range

~~(i=0; i<=n; i++)~~

Loops Definition

Loops are used to execute a block of code *multiple times*.

Range Definition

The **range function** is used to generate a sequence of numbers.

while

for

while cond:
....

for item in set:
....



Loops and Range

Loops Definition

Loops are used to execute a block of code *multiple times*.

Range Definition

The **range** function is used to generate a sequence of numbers.

$\text{range}(\text{stop})$: $(0, 1, 2, \dots, \text{stop}-1)$

$\text{range}(\text{start}, \text{stop})$: $\text{start}, \text{start}+1, \dots, \text{stop}-1$

$\text{range}(\text{start}, \text{stop}, \text{int})$:
 $\text{start}, \text{start}+\text{int}, \dots, \text{stop}-1$

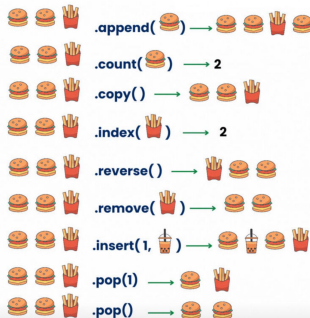


Lists

Definition

A **list** is a collection of items that are *ordered* and *changeable*. **Lists** are defined by enclosing the items in square brackets.

PYTHON LIST METHODS



Dictionaries

Definition

A **dictionary** is a collection of items that are *unordered*, *changeable*, and *indexed*. Dictionaries are defined by enclosing the items in curly braces.



Sets and Tuples

Definition Sets

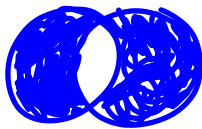
A set is a collection of items that are unordered and unindexed. **Sets** are defined by enclosing the items in curly braces.

Definition Tuples

A tuple is a collection of items that are ordered and unchangeable. **Tuples** are defined by enclosing the items in parentheses.

$A = \{a, b, c, d, \dots, z\}$ ^{unique} set

$\{a:1, b:2, \dots\}$ dict



$A \cap B$



Sets and Tuples

Definition Sets

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Definition Tuples

A **tuple** is a collection of items that are *ordered* and *unchangeable*. **Tuples** are defined by enclosing the items in parentheses.

$tuple = (a, b, c)$
 $tuple = (a,)$



Lists Comprehensions

Definition

List comprehensions provide a concise way to create lists. Common applications are to make *new lists* where each element is the result of some operation applied to each member of another sequence or iterable, or to create a subsequence of those elements that satisfy a certain condition.

$[oper(i) \text{ for } i \text{ in range}(n)]$

$[oper(i_0), oper(i_1), \dots, oper(i_n)]$

$l = []$
 for i in $range(n)$:
 $l.append(oper(i))$



Functions

Define f as a function...

Definition

A **function** is a block of code that only runs when it is called. You can pass data, known as parameters, into a function. A **function** can return data as a result.

def f(args):

... ~

~

~

return..



Functions

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A **function** is a block of code that only runs when it is called. You can pass data, known as parameters, into a function. A **function** can return data as a result.

Type of Functions

- **Built-in** Functions → *native*
- **User-defined** Functions
 - **Variadic** Functions
 - **Recursive** Functions



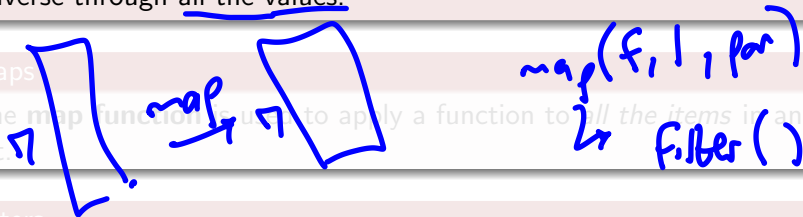
Python Iterators

Definition

An **iterator** is an object that contains a countable number of values. An **iterator** is an object that can be iterated upon, meaning that you can traverse through all the values.

Maps

The **map function** is used to apply a function to *all the items* in an input list.



Filters

The **filter function** is used to *select items* from an input list that meet a certain condition.



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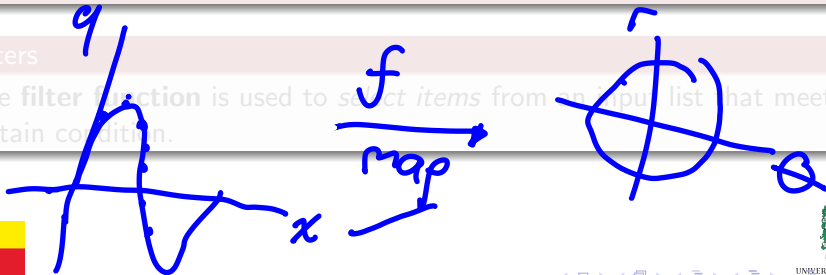
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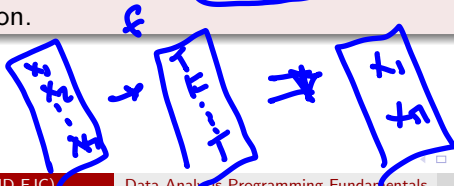
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Lambda Functions

Definition

A lambda function is a small anonymous function. A lambda function can take any number of arguments, but can only have one expression.

```
def sum(a,b):
    return a+b
```

$\lambda a,b: a+b$

```
def p3-to-str(x):
    return str(x**3)
```

$\lambda x: str(x**3)$

```
def get_isbn(book):
    return book['isbn']
```

$\lambda book: book['isbn']$



Classes and Objects

Definition

Python is an *object-oriented programming* language. Almost everything in Python is an **object**, with its *properties* and *methods*. A **class** is like an object constructor, or a “blueprint” for creating objects.



Outline

- 1 Foundations of Python
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Numerical Python Library — Numpy

- **Numpy** is the **core** library for **scientific computing** in Python. It is the **fundamental package** for scientific computing with **Python**.
- **Numpy** is a general-purpose **array-processing** package. It provides a **high-performance** multidimensional array object, and tools for working with these arrays.
- **Numpy** was created by **Travis Oliphant** in 2005, and it is an **open-source project**. Currently, **Numpy version 2.4** is available.



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Lineal Algebra with Numpy

- **Numpy** provides a comprehensive set of linear algebra functions
- **Numpy** provides the functionality to create and manipulate matrices.
- Numpy provides the functionality to solve linear systems of equations
- Numpy provides the functionality to calculate the determinant of a matrix.
- Numpy provides the functionality to calculate the inverse of a matrix.



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• **Numpy** provides the **functionality** to **calculate the determinant** of a matrix.

• **Numpy** provides the **functionality** to **calculate the inverse** of a matrix.

$$\begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$



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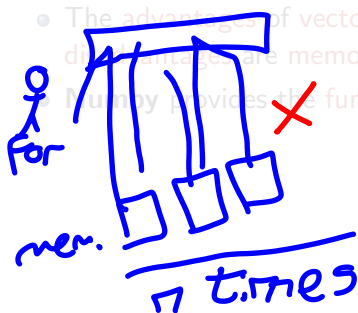
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Vectorization with Numpy

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- **Vectorization** is the process of replacing explicit loops with array expressions or matrix operations.



Vectorization with Numpy

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- **Vectorization** is the process of replacing explicit loops with array expressions or matrix operations.
- The advantages of vectorization are speed and clarity. The disadvantages are memory and complexity.
- **Numpy** provides the functionality to vectorize operations on arrays.



Typical Operations with Numpy

- **Numpy** provides the functionality to create and manipulate arrays.
- **Numpy** provides the functionality to perform element-wise operations on arrays. matrix operations on arrays.
- **Numpy** provides the functionality to perform linear algebra operations on arrays.
- **Numpy** provides the functionality to perform statistical operations on arrays.

↓
descriptive



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Introduction to Pandas

- 7p.
- **Pandas** is a fast, powerful, flexible, and easy-to-use open-source data manipulation and data analysis library built on top of the Python programming language.
 - **Pandas** is a high-level data manipulation tool developed by Wes McKinney in 2008.

• Pandas is a fast and efficient data manipulation tool that is built on top of NumPy.

• Pandas is one of the most popular and widely-used data manipulation libraries in the world.

Excel
Python



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The "Series" Data Structure

- A **Series** is a one-dimensional array-like object that contains a sequence of values and an associated array of data labels, called the index.
- The index of a **Series** is an array of labels that correspond to the values in the **Series**. The index of a **Series** is an optional parameter that defaults to a sequence of integers starting at zero.
- The **Series** object is a core data structure in **Pandas**.

Series

a_0
 a_1
 a_2
 $\vdots$
 a_n

index
 "first" 0
 "second" 1
 "third" 2
 $\vdots$
 "n-th" n

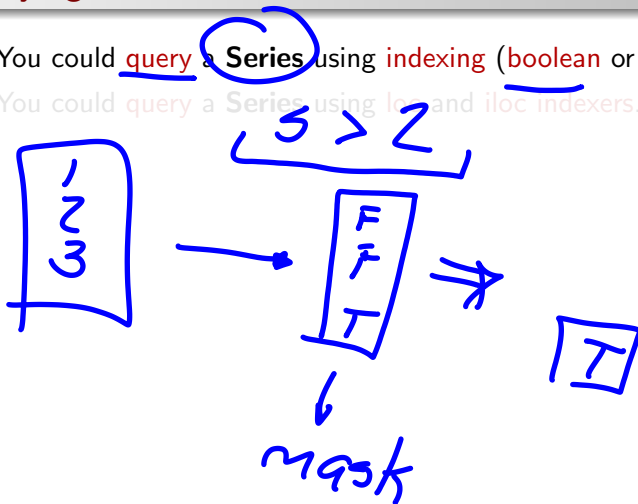
Array
 index 0 1 2 ... n
 $[a_0, a_1, \dots, a_n]$

NaN: Not a Number
 Float



Querying a Series

- You could query a Series using indexing (boolean or fancy).
- You could query a Series using loc and iloc indexes.



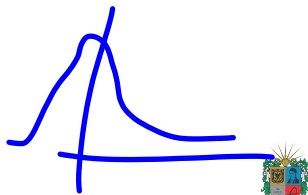
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Handwritten table illustrating indexing:

	index	value
0	a	12
1	b	24
2	c	36

not visible $\Rightarrow$ **iloc**



The "DataFrame" Data Structure

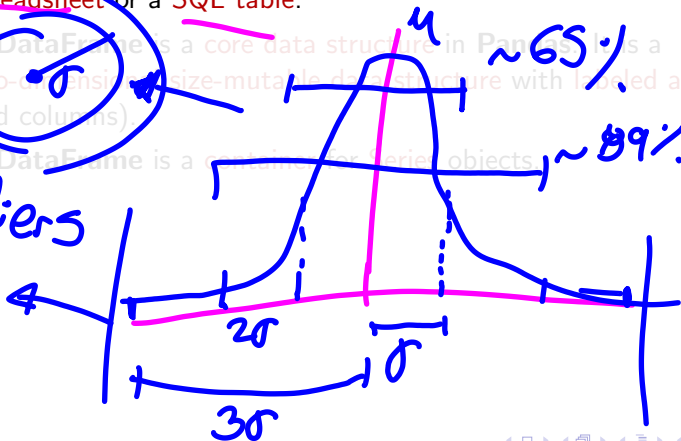
- A **DataFrame** is a two-dimensional labeled data structure with columns of potentially different types.

- A **DataFrame** is a tabular data structure that is similar to a spreadsheet or a SQL table.

- A **DataFrame** is a core data structure in Pandas. It is a two-dimensional size-mutable data structure with labeled axes (rows and columns).

- A **DataFrame** is a container for Series objects.

outliers



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DataFrame Indexing and Loading

- You could **index** a **DataFrame** using **column names**.
- You could load a DataFrame from a CSV file.
- You could load a DataFrame from a JSON file.
- You could load a DataFrame from a SQL database.

② values



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DateTime Handling in Pandas

- You could **convert** a **string** to a **datetime** object using the **to_datetime()** method.
- You could **convert** a **datetime** object to a **string** using the **strftime()** method.
- You could **convert** a **datetime** object to a **timestamp** using the **timestamp()** method.



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- You could **query** a **DataFrame** using **query** method.



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Missing Values in a DataFrame

- You could **detect** missing values in a **DataFrame**. The **isnull()** method returns a **Boolean DataFrame** indicating the **presence** of missing values.
- You could **fill** missing values in a **DataFrame**. The **fillna()** method returns a **DataFrame** with missing values filled.
- You could **drop** missing values in a **DataFrame**. The **dropna()** method returns a **DataFrame** with missing values dropped.



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Outline

- 1 Foundations of Python
- 2 Numerical Analysis with Numpy
- 3 Data Manipulation with Pandas



Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/data-analysis-programming>

